

Executive Business Report

Sales Forecasting & Demand Segmentation System

Prepared for: Head of Supply Chain & Chief Financial Officer (CFO)

1. Executive Summary

This project analyzed historical retail sales data to understand sales trends, forecast future demand, identify unusual sales patterns, and segment products based on demand behavior. SARIMA, Facebook Prophet, and XGBoost were evaluated. Facebook Prophet achieved the best forecasting accuracy. An interactive Streamlit dashboard was developed to support business decision-making.

2. Key Findings

- Sales increased steadily from 2015 to 2018.
- Technology and Furniture generated the highest revenue.
- Monthly sales exhibited seasonal variation.
- Facebook Prophet delivered the best forecasting performance.

Model	MAE	RMSE	MAPE
SARIMA	19730.43	24677.39	19.67%
Facebook Prophet	13434.28	14049.75	14.64%
XGBoost	36181.84	41854.21	36.84%

3. Three-Month Sales Forecast

Month	Forecast	Confidence Range
Month 1	61,591	56,000–67,000
Month 2	101,136	93,000–111,000
Month 3	90,674	84,000–105,000

4. Top Anomalies Detected

1. High sales spike (late 2018) – likely due to festive promotions.
2. Low sales (early 2015) – possible seasonal slowdown or inventory shortage.
3. Sales surge (early 2018) – likely marketing campaign or bulk purchase.

5. Product Demand Segmentation

- High Volume Core Products: Maintain higher inventory.
- Growing High-Value Products: Increase procurement gradually.
- Low to Medium Volume Stable Demand: Maintain regular stock levels.

6. Business Recommendations

1. Adopt Facebook Prophet for production forecasting.
2. Prioritize inventory for Phones, Chairs, Storage and Tables.
3. Monitor anomalies regularly to respond quickly to unexpected demand.

7. Risk / Limitation

Forecasts rely on historical sales data and may be affected by unexpected economic events, promotions, or supply chain disruptions.

8. Conclusion

The developed forecasting and analytics system provides valuable support for inventory planning, sales forecasting, and business decision-making. Facebook Prophet is recommended as the primary forecasting model.